



# MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

**BITCOIN AND CRYPTOCURRENCIES [IT-4423]**

Marks: 50

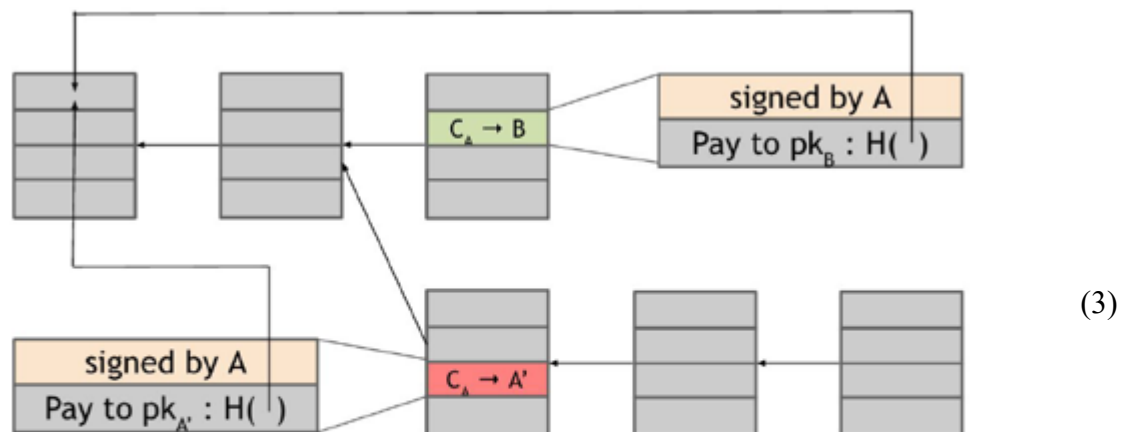
Duration: 180 mins.

## PART-A

Answer all the questions.

- 1) The Ministry of Education has launched a nationwide online certificate issuance system for graduates. Each certificate is issued in PDF format and includes a digital signature from the authorized university registrar. Students can download their certificates and share them with employers or institutions for verification. One day, an employer raises a concern: two certificates from different candidates have identical registrar signatures, but the contents (names and scores) are different. Analyse how digital signatures work in this scenario to ensure the authenticity and integrity of issued certificates. (5)

B)



Utilize double spend attack theory to describe the scenario in above diagram.

- C) Illustrate the process how Bitcoin Achieves Decentralization (2)
- 2) Raj stored 3 BTC in a non-custodial software wallet on his phone. After a factory reset, the app was deleted, and he cannot access his wallet anymore. He never noted down the seed phrase. (5)
- A) 1. Select the type of Bitcoin wallet was Raj using, and what responsibilities come with using such a wallet? (5)  
2. Identify the reason seed phrase critical in this situation?
- B) Evaluate how bitcoin script plays a role in Bitcoin transactions and describe how it enables features like multi-signature wallets and time-locked transactions. Provide a (3)

basic example of a Bitcoin Script used for a Pay-to-Public-Key-Hash (P2PKH) transaction.

- C) Describe the role of ASICs in Bitcoin mining, and the reason they are preferred over GPUs and CPUs? (2)
- 3) Draw Merkle tree data structure to build bitcoin network (5)
- A)
- B) As Bitcoin's mining difficulty adjusts approximately every two weeks based on total network hash rate, consider a scenario where a sudden influx of highly efficient ASIC miners dramatically increases the network hash rate. Analyse how this change affects block discovery time, mining difficulty, and profitability for existing miners using older hardware. (3)
- C) Describe the pseudocode to find cost of mining (2)
- 4) As a blockchain researcher, Analyse how would you assess the feasibility of transitioning Bitcoin to an alternative consensus mechanism?
- A)
- What technical, social, and governance challenges would arise from such a shift?
  - How would such a change affect Bitcoin's decentralization, security, and miner incentives? (5)
- B) Illustrate with suitable example Should the Bitcoin community embrace major protocol changes (e.g., adding smart contract capabilities or switching consensus)? What risks and benefits would come with that? (3)
- C) Classify the key factors that could lead to the collapse or destruction of the Bitcoin network? (2)
- 5) Determine the key challenges and potential benefits of adopting cryptocurrencies for everyday use by the public and explain how factors like usability, regulation, and security influence mass adoption. (5)
- A)
- B) Examine what could potentially destroy the value or existence of Bitcoin? (3)
- C) Describe the process of Allocating altcoins via proof-of-burn using suitable diagram (2)

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